

MSME IDEA HACKATHON 5.0Reference No. :- **INC25ETN087014**

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6. Theme	Stealth, Surveillance, and Cyber Defense Technologies
7. Idea Sector	Power, Renewables, Electricals, Power Electronics, Energy Efficiency and any related sub-sector
8. Title of proposed idea/innovation	Project Vyomketra
9. Briefly explain newness/uniqueness of the innovation	This innovation introduces a fully autonomous orbital strike platform that masquerades as a satellite but functions as a rapid-response military asset. Unlike traditional ballistic systems or satellite-based surveillance, this system uniquely combines: • Stealth weaponization in Low Earth Orbit (LEO) • Passive glide-based missile reentry with grid fins (undetectable by conventional radars) • Modular warhead deployment with adaptable payloads (TNT, micro-missiles, kinetic rods, etc.) • No ground-based reactivation needed — platforms are long-term autonomous • Multi-role use: surveillance, ASAT defense, and strategic retaliation — all from a single mobile platform It is a next-gen hybrid of satellite and hypersonic strike tech, optimized for space-based dominance without provocation.
10. Concept & Objective	Concept: Project Vyomk?etra envisions a network of Indian-made autonomous orbital platforms stationed in Low Earth Orbit (LEO), disguised as satellites but secretly armed with kinetic glide strike vehicles. These platforms orbit silently, anchored over India, capable of launching stealth-based atmospheric reentry missiles during wartime — all without external command or mid-orbit maintenance. Each platform is a self-sustained military “watchtower” that can survive in orbit for years, holding 20–25 precision warheads, capable of instant activation, region-targeted descent, and untraceable entry paths, beyond the reach of enemy missile defense systems. ? Objective: • To establish space-based strategic superiority over hostile regions • To provide India with a first-response orbital deterrent without ground dependency • To integrate multi-domain defense (ASAT, surveillance, missile strike) into a single unified platform • To ensure quick-launch strike readiness from space, bypassing international radar detection • To enforce psychological deterrence through a persistent, invisible orbital threat over enemy territories

11. Specify the potential areas of application in industry/market in brief	Project Vyomk?etra, though defense-centric, opens pathways for multiple strategic and industrial applications: 1. Defense Aerospace: Direct application in national security through space-based deterrence, ASAT operations, and next-gen missile deployment. It advances India's orbital combat readiness and autonomous warfare capabilities. 2. Satellite Technologies: Innovations in stealth platform design, long-duration orbital power systems, and autonomous self-regulating AI can be adapted for dual-use civilian and military satellites. 3. Space Surveillance Reconnaissance: High-precision optical, thermal, and RF sensors used on platforms can be repurposed for global monitoring, border security, and early-warning systems. 4. Hypersonic Vehicle Development: The kinetic glide missile design supports industries working on hypersonic delivery systems and atmospheric reentry tech. 5. Space Propulsion Power Systems: Nuclear microreactors and solar hybrid systems on the platform offer future pathways for sustainable deep-space exploration or high-endurance satellites.
12. Briefly provide the market potential of idea/innovation	Project Vyomk?etra targets a fast-growing global market centered on space-based defense technologies, including autonomous orbital platforms, hypersonic glide systems, and ASAT weaponry. As global powers shift military infrastructure into Low Earth Orbit, the demand for stealth-based, rapid-response systems is accelerating. India's entry into this domain secures both strategic leverage and early market advantage. The innovation supports commercialization through dual-use technologies like autonomous control systems, advanced surveillance payloads, reentry tech, and compact power units (solar-nuclear hybrids). These components have scalable applications across the defense, aerospace, and satellite industries — encouraging collaboration between DRDO, ISRO, startups, and private manufacturers. With global defense spending on space projected to cross 100 billion by 2035, Vyomk?etra's modular architecture allows India to become a defense exporter of orbital platforms, strike components, and surveillance tech. It creates an ecosystem that aligns with India's goals of self-reliance (Aatmanirbharta) and leadership in space defense innovation.
13. Uploaded Proposal	View/Download